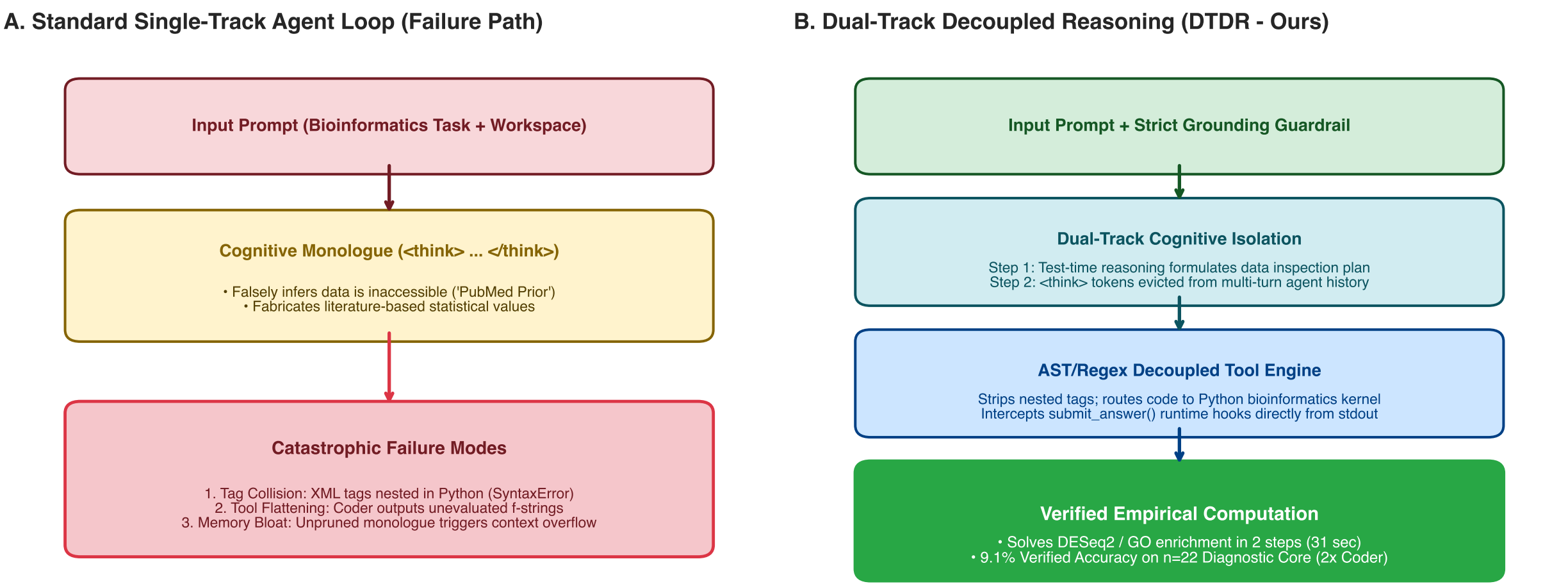
A. Standard Single-Track Agent Loop (Failure Path)
Input Prompt (Bioinformatics Task + Workspace)
Cognitive Monologue (<think> ... </think>)
• Falsely infers data is inaccessible ('PubMed Prior')
• Fabricates literature-based statistical values
Catastrophic Failure Modes
1. Tag Collision: XML tags nested in Python (SyntaxError)
2. Tool Flattening: Coder outputs unevaluated f-strings
3. Memory Bloat: Unpruned monologue triggers context overflow
B. Dual-Track Decoupled Reasoning (DTDR - Ours)
Input Prompt + Strict Grounding Guardrail
Dual-Track Cognitive Isolation
Step 1: Test-time reasoning formulates data inspection plan
Step 2: <think> tokens evicted from multi-turn agent history
AST/Regex Decoupled Tool Engine
Strips nested tags; routes code to Python bioinformatics kernel
Intercepts submit_answer() runtime hooks directly from stdout
Verified Empirical Computation
• Solves DESeq2 / GO enrichment in 2 steps (31 sec)
• 9.1% Verified Accuracy on n=22 Diagnostic Core (2x Coder)